

- 6 Two parallel metal plates X and Y are separated by a distance of 0.041 m, as shown in Fig. 6.1.

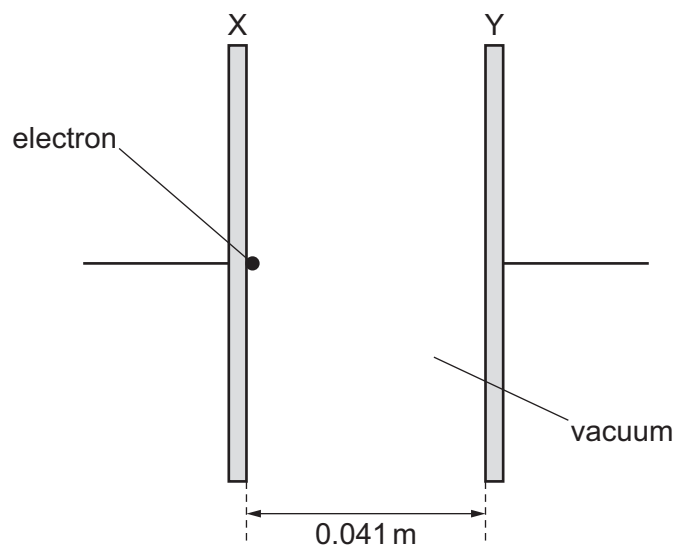


Fig. 6.1

There is a vacuum between the plates. An electron is at rest at the centre of plate X.

A potential difference (p.d.) of 58 kV is applied across the plates. This causes the electron to accelerate towards plate Y.

- (a) On Fig. 6.1, use the symbols + and – to indicate which of plates X and Y is the positive plate and which is the negative plate. [1]
- (b) (i) Calculate the electric field strength E between the plates. Give a unit with your answer.

$$E = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (ii) Determine the acceleration of the electron.

$$\text{acceleration} = \dots\dots\dots \text{ms}^{-2} [2]$$

